

# Bio-Design and Manufacturing

## Technology-Driven Solutions for Autism support: Enhancing Intervention and Inclusion --Manuscript Draft--

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# Technology-Driven Solutions for Autism support: Enhancing Intervention and Inclusion

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**Abstract** – This paper examines the need for providing individuals with autism spectrum disorder with adequate support systems and the use of technology for increasing the effectiveness of such interventions as well as fostering social inclusion is examined. An autism spectrum disorder refers to a kind of neurodevelopmental disability causing inability to communicate and behavior problems. There exist various ways of overcoming problems related to the development of an autism spectrum disorder, among which are application programs, artificial intelligence, virtual learning spaces, communication improvement measures, and monitoring systems. Due to their implementation, specialists are able to provide better opportunities for making learning easier, communicating, and being monitored in order to offer personalized assistance. Moreover, through the implementation of new technologies, favorable conditions are created for ensuring involvement in certain activities for people with autism in order for them to be able to interact in schools or places of employment. According to empirical results, the implementation of both technological advances and conventional measures of assistance ensures more efficient access to learning opportunities and achievement of desired results. In this context, technology that is flexible and easy to use to help customize the service, thus ensuring that technologies that are used in treating autism can be accessed, becomes necessary.

**KEYWORDS-** AUTISM SPECTRUM DISORDER, ASSISTIVE TECHNOLOGY, ARTIFICIAL INTELLIGENCE, VIRTUAL REALITY, DIGITAL INTERVENTION, INCLUSIVE EDUCATION, EARLY INTERVENTION, TECHNOLOGY AND HEALTHCARE

## I. INTRODUCTION

The knowledge gained about neurodevelopmental disorders results in the need for assistance systems that attempt to improve the lives of those affected by the illness. Autism is among the illnesses with elements including speech, social interactions, and repetitive behaviors. According to the information from various health institutions across the globe, there is one child out of every 100 children who has autism.

The traditional models of therapy for autism are associated with the usage of behavioral therapy, speech therapy, and special education programs. Undoubtedly, all these models play a crucial role in the improvement of the state of patients suffering from the disease; yet, they are rather complicated, expensive, and require a highly professional approach. Thus, most people cannot benefit from the above-mentioned treatment options because of their high cost, especially in developing countries. In other words, there were many individuals whose lives could change drastically under the influence of the described therapy.

However, the application of technology could provide an appropriate solution to the discussed problem. There is a variety of models of technological treatment, such as apps, artificial intelligence, virtual reality, wearable devices, and communication. This paper identifies different technology-based solutions that could aid individuals suffering from autism. The paper also looks at the potential benefits of such solutions in enhancing the inclusion approaches. Technology-based solutions can be extremely effective in autism intervention measures together with other methods.

However, there has been no comprehensive study that compares all the main technological methods used in ASD treatment in one context. The current paper seeks to address this issue by making a systematic comparison between AI, mobile applications, VR, and wearable technologies to find the best means of addressing ASD.

## II. LITERATURE REVIEW

The recent research has greatly expanded the opportunities for the employment of computer simulation models in the process of ASD screening and identifying its severity levels. Despite the success of these researches in achieving their objectives, the aim of this research is to build upon these recent researches by ensuring the use of multiple input data sources.

To cite an example, the study carried out by Robles et al. [1] has shown the use of VR-based system, which uses embodied agents, to analyze non-verbal responses. Using machine learning models, such as SVM and LSTM, for analyzing eye-gaze and head movement of participants, Robles et al. [1] have managed to achieve a maximum accuracy of 100%. On the other hand, Hassan et al. [2] have also used RET-based framework employing the concept of Adaptive Boosted 3D biomarkers for analyzing ASD level in participants. Hassan et al. [2] have succeeded in achieving a maximum accuracy of 95.59% in their research.

Deep learning technology can also be employed successfully in analyzing biological and behavioral data. As shown in the work of Eni et al. [3], a Deep Neural Network algorithm was applied in the research in order to estimate the level of autism, analyzing the patterns of vocal signal from speech samples. A correlation coefficient of 0.72 obtained between the results of CNN algorithm and clinical parameters obtained from ADOS score makes one conclude that computer-assisted vocal behavior estimation is quite possible as a substitute for more conventional methods. The analysis of brain, performed by Zhang et al. [4], included estimating its functional connectivity using EEG features and detecting patterns of activity in the frontal and occipital lobes.

In addition, predictive models could also be used in calculating the probabilities of certain diseases in relation to genetics. For instance, a good example is the predictive modeling done by Carvalho et al., whereby HMM model was employed in calculating the probability of developing autism spectrum disorder in accordance with the genetic background of the family [5]. This has been instrumental in providing a better understanding of the risks involved, the probability of which can range from 0% up to 80%. On the contrary, the model known as "Autism Assistant", developed by Chistol et al. was a way of providing an intervention that integrates both the therapeutic and home-based intervention in a human-centered manner [6]. Although the model works efficiently through the administration of ABA therapy, the main purpose is to train parents to become co-therapists.

### A. Mobile Applications and AAC Tools

As per Bhatt et al. [7], there had been some studies on the usage of communication aids through mobile applications for minimally verbal autistic children. Based on the findings of the study, it could be seen that the pictorial or symbol-based applications improved expressive communication skills of the children when used in treatment sessions and even at home. Besides, it was noted that caregivers had played an essential role in helping achieve success in such cases. Furthermore, it was stated that the highest performing applications were the ones that featured audio and visual aid tools.

The research conducted by Parsons et al. [8] focused on the involvement of children diagnosed with ASD in technological interventions. Findings of this research suggest that a controlled technological environment helps to maintain children's attention and avoid behavioral problems. In particular, it was stated that children involved in learning through technology showed higher levels of attention than those children attending traditional classes. Moreover, according to Parsons et al., individualized display of information, possible via mobile apps, is a crucial factor when maintaining attention of autistic children.

### B. Wearable Devices and Physiological Monitoring

In the research conducted by Voss et al. [9], attention was paid to the possibilities that exist for physiological monitoring of children diagnosed with ASD using wearable sensors. As a result, based on the physiological monitoring provided through continuous heart rate variability monitoring, skin conductance assessment, and movement pattern analysis, the research concluded that prediction of behavioral dysregulation becomes possible through analysis of the data obtained through physiological monitoring systems. Using machine learning technologies for predicting the meltdowns with the help of sensor data, proactive intervention of the behavior became possible. Thus, the research demonstrates that wearable sensors possess considerable potential for regular intervention in behavior.

As for Goodwin et al.'s research [19], the paper could be regarded as one of the earliest dedicated to the problem of autism. The main goal of the research was to identify any differences in the cardiovascular arousal in patients suffering from autism compared to the control group, and the research proved that differences existed in reaction to the stimuli. Thus, the physiological patterns that characterize the subject became identified.

### C. Serious Games and Educational Technology

Khowaja and Banire [10] provide a thorough literature review on the use of serious games in helping children with autism spectrum disorder (ASD) to acquire learning skills. As per the findings of Khowaja and Banire [10], who studied 84 published papers within the years 2018-2024, the implementation of serious games proved to be effective in enhancing the learning abilities related to reading, writing, numeracy skills, and emotional recognition. What is more critical is that the researchers observed higher efficacy when it came to games that were specifically created taking into account the peculiarities of autistic patients compared to those adapted for their needs.

Nonetheless, one of the earliest studies conducted by Zakari et al. [20] focused on developing serious games intended exclusively for the use of children with ASD. As per their findings, serious games could be used for addressing various problems faced by children, such as social cognition, turn-taking, and facial affect recognition ability. Importantly, the involvement of occupational therapists and special educators in designing video games was noted as a vital component of the research conducted by the authors. Nevertheless, an improvement in evaluation techniques was needed because of the small sample size utilized in the experiment.

#### **D. AI-Based Facial Analysis and Eye Tracking**

Bi et al. [11] introduced an innovative machine learning system using multimodal methods whereby it was proposed that there should be a screening mechanism for autistic spectrum disorder patients based on the recognition of facial expressions. According to the study by Bi et al. [11], facial micro expressions and eye tracking were successfully identified using both CNN and SVM methods with an accuracy level comparable to that of expert ratings. As noted by Bi et al. [11], facial expressions can work as a potential screening tool when expertise is not available.

Mundy et al. [15] provided an overview of eye-tracking devices as biomarkers for autism spectrum disorder, summarizing the findings from several research studies that investigated eye movements in response to social and nonsocial stimuli. Mundy et al. concluded that abnormal gaze behavior – such as difficulty in focusing on faces, joint attention, and impaired saccade movements – is an accurate predictor of ASD and could be observed as early as six months in infancy by employing commercial eye-tracking equipment. The authors suggested the need for standardized procedures of eye-tracking technology to make cross-comparisons possible across various studies and highlighted the potential of eye tracking as an affordable diagnostic tool.

#### **E. Social Robots and Augmented Reality**

Speaking of the research conducted by Alcorn et al. [13], which was dedicated to the utilization of social robots in communication therapy for autistic children, it should be said that there was information regarding the advantages and disadvantages of using social robots proposed by different scholars in the scientific domain. In fact, it was found that unlike humans, robots could be considered better communicators when dealing with autistic children because they have an objective attitude toward communicating with humans. Additionally, it was revealed that autistic children readily participated in turn-taking and eye-tracking tasks.

In order to investigate the effectiveness of using AR technology in developing social skills of ASD children, an experimental study was conducted by Tariq et al. [14]. For this study, an AR system was built in which interactive social activities were overlaid on reality and allowed the children to develop their social skills in a suitable environment. The outcomes showed that following the eight-week therapy program, there was a remarkable change in the social responsiveness scores of the subjects with better skill generalization than the use of VR.

#### **F. Facial Affect Recognition and Parent-Mediated Interventions**

Another study was conducted by [18] Tanaka et al., where computer-based interactive learning was applied as an approach to training individuals with ASD in facial emotion recognition. The application of animated faces played a key role in making it possible for people to learn facial emotion recognition. In this particular study, it was discovered that there were significant improvements in relation to the identification of basic emotions such as happiness and sadness, and transfer occurred during face identification. It can be noted that computer-based affect

training is indeed quite effective as a relatively inexpensive treatment approach.

Kasari et al. [16] performed a randomized comparative study on the efficacy of parent-based interventions in toddlers with autism. The researchers found that interventions administered by trained parents were as effective as those provided by clinicians in fostering joint attention and play skills among autistic toddlers. This is an important insight when developing technology-based systems, since it supports the design approach of empowering caregivers with technological tools like behavior tracking and intervention assistance applications to ensure therapeutic effects outside of the clinical setting.

From the above discussion, it can be seen that there is always a distinctive way in which each of the technologies discussed above from mobile application technology to social robotics addresses an intervention requirement amongst individuals with autism. However, the intervention techniques among individuals with autism spectrum disorder cannot be tackled with any of the technologies discussed above because there is no effective integrated intervention approach. Integration of all the intervention approaches would yield significant success.

### **III. METHODOLOGY**

#### **A. Dataset**

The aim of this research project is to examine how the use of new technologies can assist people diagnosed with autistic spectrum disorders. In the past, researchers have performed studies using one data set of number calculations; currently, the authors conducted an analysis and compare multiple sources (literature) that relate to different types of technology that have been developed to assist people who are diagnosed as being on the autism spectrum. Multiple sources (papers, abstracts presented at conferences) and reports published by health care facilities were reviewed regarding the multiple forms of technology developed to help individuals on the autism spectrum. The research was based on several high-quality electronic databases (e.g., Google Scholar ([www.scholar.google.com](http://www.scholar.google.com)), PubMed ([www.pubmed.org](http://www.pubmed.org)), WHO ([www.who.int](http://www.who.int))), which all provide reliable information in the form of scientific articles and research studies related to assistive technologies and autism. This research used the data collected from reviewing the electronic database as the basis for creating a comprehensive overview of how to use modern technology to assist those with autism.

#### **B. Attribute Description :**

The characteristics of technology are the essential components for individuals with Autism and their support systems, as identified in this study. Furthermore, the characteristics of technology can be viewed in terms of the capabilities of technology and most of the capabilities can assist autistic individuals in resolving many of the challenges they face. A characteristic of interest in technology is communication support. Many individuals with autism experience difficulty communicating with others in a manner that expresses their desires and emotions. There are many types of communication tools available to assist those with autism to communicate with others through symbols or written text. The

reason that the characteristics of technology are important is because many individuals with autism have difficulty communicating effectively through speech.

An important aspect of this technology is its ability to assist in social interaction training. Research shows that there are difficulties encountered by those individuals with autism when it comes to perceiving identity and recognizing both facial expressions and interpreting social interactions. One way to support those individuals living on the spectrum is with virtual reality technology to provide individuals who have been diagnosed with autism a training environment to work on social interaction. Such a major component of the technology-based approach to social interaction is to provide those diagnosed with autism an opportunity to engage with others in a socially supported manner. Additionally, the types of technologies available for individuals on the spectrum also include ways to monitor their behaviors. There are numerous wearable technologies on the market with sensors designed to measure a variety of physical reactions including but not limited to an individual's heart rate, movement, and level of stress. This type of monitoring can help provide an early response to the individual on the spectrum. Furthermore, technology can provide individuals with the necessary tools to develop an effective and positive learning experience through the use of visual instructions, games, and an increase in levels of engagement and focus for individuals diagnosed with autism. Currently, many of these tools are being implemented as part of the learning environment and special education settings.

The earlier part of the discussion included an element regarding the ability of new technology (AI-enabled) to provide diagnosis regarding autism through the early detection of the disorder via technology. Such a system would employ AI-based diagnostics that could identify autism early through monitoring behavior, analyzing facial expressions, and tracking eye movements. The early detection of the disorder via technology (such as the AI-enabled tool) will be an important area for researchers looking to provide scientific evidence for how technology can be utilized to customize treatment for individuals diagnosed with autism.

### C. Research Methodology

In this research project, qualitative research methods were used to determine how important technology is for assisting people with autism. Rather than collecting original data through primary research, existing literature (including research articles and technical reports) that has been written by a variety of authors was used as the source of information on technology. To conduct this research, the first step was to search for relevant sources of literature for analysis during the research process, specifically in terms of technology for identifying, treating, and providing therapy for autism. There are a large number of health and academic websites that contain information relevant to the subject of this research study. Once the research materials have been identified, the analysis was based on the particular types of technology that are being studied, which included artificial intelligence applications, mobile communications, wearable technology, and virtual reality-based therapy.

The comparative analysis of the research articles, provided an opportunity for a deeper understanding of the contributions

each of the technologies makes to the advancement of the various aspects related to autism intervention, including enhancements in communication, behavioral observation, social interaction and education. In addition, the strengths and weaknesses of each of the technologies were also evaluated. This information can be used to determine which technology will provide the greatest benefit to individuals who have autism spectrum disorder.

This type of analysis provided an understanding of how technology can benefit autism interventions.

### D. Classifiers

In order to analyze these types of technological interventions in detail, researchers have grouped similar types of technology into categories according to how they function and what type of technology they are. The first group of technological interventions discussed in this research paper is made up of AI technologies. AI technologies use machine learning to analyze a variety of behavioral, developmental, and facial recognition characteristics to be able to provide healthcare professionals with a means of identifying patients who have autism in the very early stages of development. The second type of digital tool is represented by mobile app technology. In fact, there are numerous mobile app technologies that are specifically designed to assist individuals with autism in communicating more effectively with others. For example, there are many mobile apps that assist individuals with autism in better communicating with others using a combination of both visual and auditory stimuli.

A virtual reality system is an alternative source used, as it produces a space for autism individuals to simulate interactions in the classroom, with peers, and with strangers.

Lastly, to use a form of technology known as "wearable devices". These devices have sensors that record and transmit a variety of signals emitted by the user (e.g. - a user with ASD).

### E. Performance Classifiers

To determine how effective the technology-based solutions that have been suggested to address the problems associated with autism are, a number of different factors need to be taken into account. First of all, the factor that has been examined in this research to give an indication of the effectiveness of technology-based solutions to autism issues was the improvement in a patient's ability to communicate after using the technology. When a patient is better at communicating, they can express their needs and other things of importance to them more clearly. Because communication is such an important issue for individuals with autism, it can also be used to evaluate how effective the technological solution is.

Another parameter that has been evaluated to assess the effectiveness of the technological solution for autism, was whether that technological solution has helped enhance the development of the social connections for the autistic patient. Furthermore, the learning and cognitive progress of the autistic individual are also very important metrics for evaluating the effectiveness of technology-based solutions to autism.

Lastly, the efficiency of the behavioral monitoring process has also been evaluated for evaluation purposes; specifically,

there have been evaluations of how efficiencies related to wearable technology and sensor technology have improved the measuring of stress levels in individuals with autism. These parameters mentioned above have provided a clearer picture of how technology may be able to assist in improving the quality of life for persons living with autism.

#### IV. RESULTS AND DISCUSSION

The research paper focuses on comparative analysis of different technology-based interventions that have been used to assist people with Autism Spectrum Disorder (ASD). These technology-based interventions are AI-Based Intervention, Mobile App-Based Interventions, Virtual Reality Interventions, and Wearable Device-Based Interventions. The current study is based on extensive literature review, which evaluates the success of these interventions based on their performance metrics, which include improvement in communication, social skills, learning abilities, and behaviors.

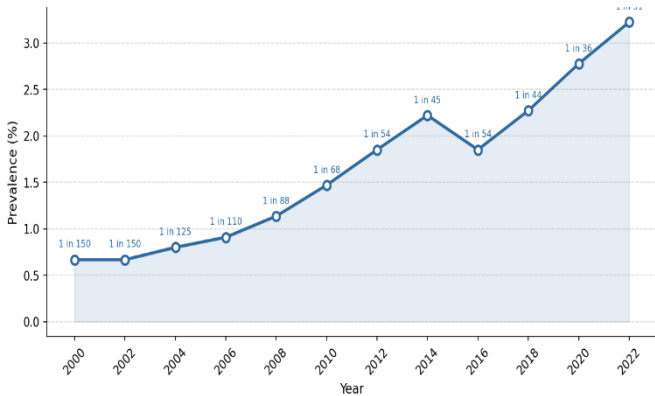


Figure 1: ASD Prevalence Among 8-Year-Old Children in the U.S. (CDC ADDM Network, 2000-2022)

In research conducted specifically for assessing ASD in infants, it has been shown that technological advances associated with AI have demonstrated considerable success when utilized to evaluate the degree and severity of ASD in its earliest stages due to the ability of the AI-based system to analyze the voice characteristics, behaviors, and physiological properties of each individual infant. This allows clinicians to make decisions regarding the appropriate level of treatment for each infant without any potential for bias.

**Mobile applications** have also demonstrated substantial benefits in assisting individuals with communication disabilities to develop effective means of communicating. This is primarily due to the flexibility and ease of use of mobile applications across both clinical and home environments. Mobile applications are considered particularly effective for this purpose because they can support a variety of functions including visual stimuli and speech generation.

The effectiveness of **VR systems** in assisting individuals to develop improved social interaction skills has also been established. VR systems allow individuals the ability to practice

their social skills in an environment that is safe and secure, thereby increasing their level of confidence and reducing levels of anxiety related to participating in these types of activities.

**Wearable devices** assist in monitoring people's behavior by monitoring their heart rates, stress levels, and movements. This helps the caregivers identify the problems at an early stage and take action to resolve them before the situation becomes worse. This leads to effective caregiving and intervention results.

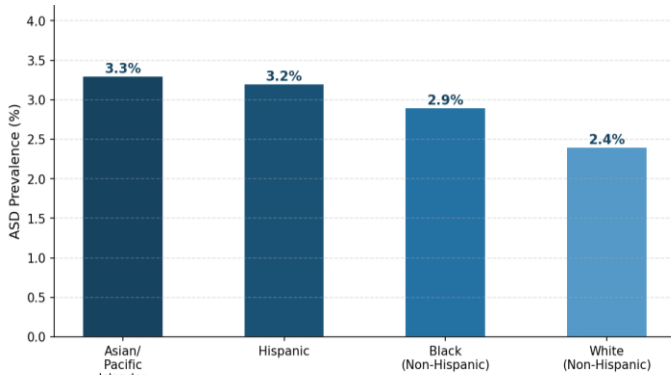


Figure 2: ASD Prevalence by Race/Ethnicity Among 8-Year-Olds (CDC ADDM Network, 2020, Published 2023)

**Table I** shows the comparison of different technologies based on evaluation parameters is represented in the table below. It is clear that AI-based systems and VR technologies are effective overall, while mobile apps are effective in being easy to use and accessible. Also, wearable devices are effective for monitoring, but not effective in interacting with other devices.

**TABLE I**  
**Comparative analysis of Technology-Based solutions for ASD.**

| Technol<br>ogy          | Communica<br>tion | Social<br>Interact<br>ion | Learn<br>ing | Behavio<br>ral<br>Monitor<br>ing |
|-------------------------|-------------------|---------------------------|--------------|----------------------------------|
| AI<br>Systems           | High              | Medium                    | High         | Medium                           |
| Mobile<br>Apps          | Very High         | Medium                    | High         | Low                              |
| VR<br>Systems           | Medium            | Very<br>High              | High         | Low                              |
| Wearabl<br>e<br>Devices | Low               | Low                       | Mediu<br>m   | Very<br>High                     |



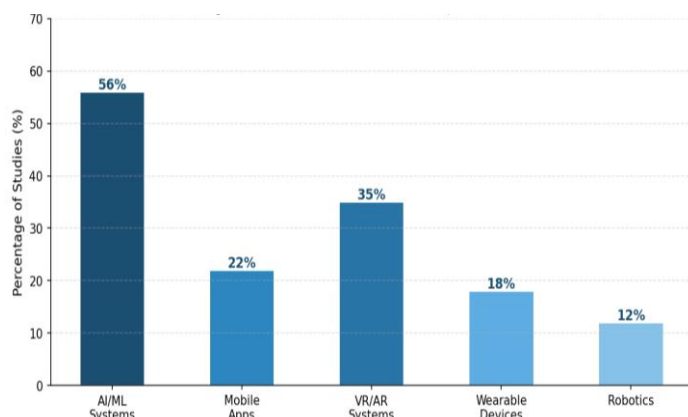


Figure 3: Distribution of Technology Categories in ASD Research (Systematic Review, 84 Studies, 2018-2024)

The results show that there is no technology that would address all the issues surrounding autism spectrum disorder. The best course of action would be to use a combination of different technologies. While different technologies offer different strengths, factors like cost, accessibility, and the requirement for technical skills continue to be a great obstacle.

## V. CONCLUSION

Through this study, there is a need to increase use of technologies by individuals on the autism spectrum through the results of this analysis. The qualitative analysis of the available literature demonstrates that there is an important role for technology, such as artificial intelligence; machine learning; virtual reality; wearable computing; and assistive communication systems, in enhancing communication modeling, behavior analysis, social interaction and cognitive development for the population with autism spectrum disorders.

The study findings provide sufficient evidence to suggest that combining intelligent systems with traditional interventions will greatly improve the efficacy of autism spectrum disorder intervention programs. Some examples of technological advances to improve quality and development for individuals with autism spectrum disorder include early diagnosis through predictive analytics; real-time behavior analysis using sensor-based technologies; and immersive learning through virtual reality.

Some of the directions that future research may pursue avenues such as creating hybrid or ensemble systems that integrate the previously mentioned technologies to create a sin Diagnostic Support System. Systems must also be made scalable, user-centered, and economically deployable in various contexts. Examples of possible future research directions include integrating real-time systems, personalizing systems using data analysis, or providing health care solutions via the cloud in order to develop an autism support system. These advancements will significantly benefit healthcare through the use of intelligent technologies that will improve the treatment of patients with autism spectrum disorder.

## CONFLICT OF INTEREST

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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